

30-DAY SQL MICRO COURSE | HOSPITAL DATA ANALYSIS

Portfolio Assignment • Hospital_Data.csv • MySQL

Student Name		Database	hospital_data
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Objective: Use SQL aggregation, grouping, sorting and date functions to extract practical hospital-data insights.

1. Total Patients

SQL: SELECT SUM(`Patients Count`) AS total_patients FROM hospital_data;

Result: 10,347

2. Average Doctors

SQL: SELECT ROUND(AVG(`Doctors Count`),2) AS average_doctors FROM hospital_data;

Result: 27.08 doctors

3. Top 3 Departments

SQL: SELECT `Department`, SUM(`Patients Count`) AS total_patients FROM hospital_data GROUP BY `Department` ORDER BY total_patients DESC LIMIT 3;

Result: Urology 1,766 | Neurology 1,229 | ENT 1,064

4. Maximum Medical Expense

SQL: SELECT `Hospital Name`, `Medical Expenses` FROM hospital_data ORDER BY `Medical Expenses` DESC LIMIT 1;

Result: Healing Touch – 49,955.41

5. Daily Average Expense

SQL: SELECT `Hospital Name`, ROUND(AVG(`Medical Expenses`/DATEDIFF(STR_TO_DATE(`Discharge Date`, '%d-%m-%Y'),STR_TO_DATE(`Admission Date`, '%d-%m-%Y'))),2) AS daily_avg FROM hospital_data GROUP BY `Hospital Name`;

Result: Highest: Fortis Care – 4,585.40/day

6. Longest Stay

SQL: SELECT `Hospital Name`, `Department`,DATEDIFF(STR_TO_DATE(`Discharge Date`, '%d-%m-%Y'),STR_TO_DATE(`Admission Date`, '%d-%m-%Y')) AS stay_days FROM hospital_data ORDER BY stay_days DESC LIMIT 1;

Result: Apollo Health, ENT – 15 days

7. Patients by City

SQL: SELECT `Location`, SUM(`Patients Count`) AS total_patients FROM hospital_data GROUP BY `Location` ORDER BY total_patients DESC;

Result: Jaipur 1,505 | Ahmedabad 1,467 | Lucknow 1,264 | Hyderabad 1,261

8. Avg. Stay by Department

SQL: SELECT `Department`, ROUND(AVG(DATEDIFF(STR_TO_DATE(`Discharge Date`, '%d-%m-%Y'),STR_TO_DATE(`Admission Date`, '%d-%m-%Y'))),2) AS avg_stay FROM hospital_data GROUP BY `Department` ORDER BY avg_stay DESC;

Result: Neurology 9.25 | Pediatrics 9.11 | Urology 8.72 days

9. Lowest-Patient Department

SQL: SELECT `Department`, SUM(`Patients Count`) AS total_patients FROM hospital_data GROUP BY `Department` ORDER BY total_patients ASC LIMIT 1;

Result: Cardiology – 544

10. Monthly Expenses

SQL: SELECT DATE_FORMAT(STR_TO_DATE(`Admission Date`, '%d-%m-%Y'), '%Y-%m') AS month, ROUND(SUM(`Medical Expenses`),2) AS total_expenses FROM hospital_data GROUP BY month ORDER BY month;

Result: Highest: Sep 2023 – 341,284.23

Conclusion: This project demonstrates practical SQL skills in data aggregation, analysis and date-based reporting—suitable for a SQL/Data Analytics portfolio.